

Produit : Electronic / Measuring devices

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INDIA

Product's definition

Category

In which category the product is classified ?

It is classified in the category of 'Audio, Video and Multimedia Systems and Equipment'.

It is regulated under IS 616 (2010): Audio, Video and Similar Electronic Apparatus - Safety Requirements [LITD 7: Audio, Video and Multimedia Systems and Equipment]

This product maybe governed by WPC certification guidelines.

Definition

What is the exact definition of the product ?

And if it exists, what is the interpretation of the authorities regarding it ?

No such Exact Definition.

Not Applicable.

Comments

Comments

IS 616 (2010): Audio, Video and Similar Electronic Apparatus - Safety Requirements [LITD 7: Audio, Video and Multimedia Systems and Equipment]

NATIONAL FOREWORD

This Indian Standard (Fourth Revision) which is identical with IEC 60065 : 2005 'Audio, video and similar electronic apparatus - Safety requirements' issued by the International Electrotechnical Commission (IEC) was adopted by the Bureau of Indian Standards on the recommendation of the Audio, Video and Multimedia Systems and Equipments Sectional Committee and approval of the Electronics and Information Technology Division Council.

This standard was first published in 1957 and subsequently revised in 1981, 1986 and 2003. The fourth revision of this standard is being done to bring it in line with the latest edition of IEC 60065 : 2005.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
- b) Comma (,) has been used as a decimal marker in the International Standard while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are listed below along with their degree of equivalence for the edition indicated:

International Standard	Corresponding Indian Standard	Degree of Equivalence
IEC 60027 (All parts) Letter symbols to be used in electrical technology	IS 3722 (Parts 1 and 2) : 1983 Letter symbols and signs used in electrical technology	Technically Equivalent
IEC 60068-2-6 : 1995 Environmental testing - Part 2: Tests - Fe: Vibration (sinusoidal)	IS 9001 (Part 13) : 1981 Guidance for environmental testing: Part 13 Vibration (sinusoidal) test	do
IEC 60068-2-32: 1975 Environmental testing - Part 2: Tests - Test Ed: Free fall (Procedure 2)	IS 6002 (Part 10/Sec 1) : 1985 Specification for equipment for environmental tests for electronic and electrical items: Part 10 Shock test machine, Section 1 Free fall type	do
IEC 60068-2-75 : 1997 Environmental testing - Part 2-75: Tests - Test Eh: Hammer tests	IS 9000 (Part 7/Sec 7) : 2006 Basic environmental testing procedures for electronic and electrical items: Part 7 Impact test, Section 7 Test Eh: Hammer tests	Identical
IEC 60068-2-78: 2001 Environmental testing - Part 2: Tests - Test Cab: Damp heat, steady state	IS 9000 (Part 4) : 1979 Basic environmental testing procedures for electronic and electrical items: Part 4 Damp heat (steady state)	Technically Equivalent
IEC 60085 : 2004 Thermal evaluation and classification of electrical insulation	IS 1271 : 1985 Thermal evaluation and classification of electrical insulation (first revision)	Technically Equivalent
IEC 60112 : 2003 Method for determining the comparative and the proof tracking indices of solid insulating materials under moist conditions	IS 2824 : 1975 Method for determining the comparative tracking index of solid insulating materials under moist conditions (first revision)	do
IEC 60127 (All parts) Miniature fuses	IS/IEC 60127 (All parts) Miniature fuses	do
IEC 60167 : 1964 Methods of test for the determination of the insulation resistance of solid insulating materials	IS 2259 : 1963 Methods of test for determination of insulation resistance of solid insulating materials	do

IEC 60216 (All parts) Guide for the determination of thermal endurance properties of electrical insulating materials	IS 8504 (All parts) Guide for determination of thermal endurance properties of electrical insulating materials	do
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IEC 60227 (All parts) Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V	IS 694 : 1990 PVG Indulated cables for working voltages upto and including 1 100 V (third revision)	do
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IEC 60245 (All parts) Rubber insulated cables - Rated voltages up to and including 450/750	IS 9968 (Part 1) : 1998 Specification for elastomer insulated cables: Part 1 For working voltages up to and including 1 100 V (first revision)	do
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IS 9968 (Part 2) : 2002 Specification for elastomer insulated cables: Part 2 For working voltages from 3.3 kV up to and including 3 kV (first revision)

IEC 60249-2 (All specifications) Base materials for printed circuits Part 2: Specifications	IS 5921 (All parts) Specification for metal-clad base materials for printed circuits for use in electronic and telecommunication equipment	do
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IEC 60268-1 : 1985 Sound system equipment - Part 1:General	IS 15596 (Part 1) :2005 Sound system equipment: Part 1 General	Identical
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IEC 60317 (All parts) Specifications for particular types of winding wires	IS 13730 (All parts) Specification for particular types of winding wires	Technically Equivalent
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IEC 60335-1 : 2001 Household and similar electrical appliances - Safety Part 1: General requirements	IS 302 (Part 1) : 2008 Safety of household and similar electrical appliances: Part 1 General requirements (sixth revision)	do
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IEC 60384-1 : 1999 Fixed capacitors for use in electronic equipment Part 1: Generic specification	IS 7305 : 1984 Specification for fixed capacitors used in electronic equipment (first revision)	do
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IEC 60384-14: 1993 Fixed capacitors for use in electronic equipment - Part 14: Sectional specification: Fixed capacitors for electromagnetic interference suppression and connection to the supply mains	IS QC 302400 : 1994 Fixed capacitors for use in electronic equipment sectional specification for fixed capacitors for electromagnetic interference suppression and connection to the supply mains	Technically Equivalent
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IEC 60417 (All parts) Graphical symbols for use on equipment	IS 2032 (All parts) Graphical symbols used in electrotechnology	do
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IEC 60454 (All parts) Specifications for pressure-sensitive adhesive tapes for electrical purposes	IS 7809 (All parts) Specification for pressure sensitive adhesive insulating tapes for electrical purposes	do
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IEC 60529 : 1989 Degrees of protection provided by enclosures (IP Code)	IS 12063 : 1987 Classification of degrees of protection provided by enclosures of electrical equipment	do
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IEC 60664-1 : 1992(revised in 2002) Insulation coordination for equipment within low-voltage systems - Part 1: Principles, requirements and tests	IS 15382 (Part 1) : 2003 Insulation coordination for equipment within low-voltage systems: Part 1 Principles, requirements and tests	do
IEC 60064-3 : 2003 Insulation coordination for equipment within low-voltage systems - Part 3: Use of coatings to achieve insulation coordination of printed board assemblies	IS 15382 (Part 3) : 2006 Insulation coordination for equipment within low-voltage systems: Part 3 Use of coating, potting or moulding for protection against pollution	Identical
IEC 60691 : 2002 Thermal links Requirements and application guide	IS/IEC Pub 691 : 1993 Thermal links - Requirements and application guide (first revision)	Technically Equivalent
IEC 60695-2-2 : 1991 Fire hazard testing - Part 2: Test methods-Section 2: Needle-flame test	IS 11000 (Part 2) : 1984 Fire hazard testing: Part 2 Test methods, Section 2 Needle-flame test	do
IEC 60707 : 1999 Flammability of solid non-metallic materials when exposed to flame sources - List of test methods	IS 11731 (Part 1) : 1986 Methods of test for determination of the flammability of solid electrical insulating materials when exposed to an igniting source: Part 1 Horizontal specimen method	do
	IS 11731 (Part 2) : 1986 Methods of test for determination of the flammability of solid electrical insulating materials when exposed to igniting source: Part 2 vertical specimen method	
IEC 60851-3: 1996 Methods of test for winding wires- Part 3: Mechanical properties	IS 13778 (Part 3) : 1993 Methods of test for winding wires: Part 3 Mechanical properties	do
IEC 60851-5 : 1996 Methods of test for winding wires - Part 5: Electrical properties	IS 13778 (Part 5) : 1993 Methods of test for winding wires: Part 5 Electrical properties	Technically Equivalent
IEC 60851-6 : 1996 Methods of test for winding wires - Part 6: Thermal properties	IS 13778 (part 6) : 1993 Methods of test for winding wires: Part 6 Thermal properties	do
IEC 60884 (All parts) Plugs and socket-outlets for household and similar purposes	IS 1293 : 2005 Plugs and socket outlets of rated voltage up to and including 250 volts and rated current up to and including 16 amperes Specification (third revision)	do
IEC 60950 : 1999(revised in 1998) Safety of information technology equipment	IS 13252 :2003 Information technology equipment - Safety - General requirements	do
IEC 61051-2 : 1991 Varistors for use in electronic equipment - Part 2: Sectional specification for surge suppression varistors	IS QG 420100: 1994 Varistors for use in electronic equipment - Sectional specification for surge suppression varistors	do

IEC 61260 : 1995 Electroacoustics, Octave-band and fractional-octave-band filters	IS 6964 : 2001 Electroacoustics-Octave-band and fractional-octave-band filters (first revision)	Identical
ISO 261 : 1973(revised in 1998) ISO general purpose metric screw threads - General plan	ISO 4218 (Part 2) : 2001 ISO general purpose metric screw threads: Part 2 General plan	Technically Equivalent
ISO 262 : 1973(revised in 1998) ISO general purpose metric screw threads - Selected sizes for screws, bolts and nuts	ISO 4218 (Part 4) :2001 General purpose metric screw threads: Part 4 Selected sizes for screws, bolts and nuts (second revision)	do
ISO 306 : 1994 Plastics - Thermoplastic materials - Determination of vicat softening temperature (VST)	IS 13360 (Part 6/Sec1) : 1999 Plastics - Methods of testing: Part 6 Thermal properties, Section 1 Determination of vicat softening temperature of thermoplastic materials	Identical

The technical committee has reviewed the provisions of the following International Standards referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

International/Other Standard	Title
IEC 60038 : 1983	IEC standard voltages
IEC 60086-4 : 2000	Primary batteries - Part 4: Safety of lithium batteries
IEC 60320 (All parts)	Appliance couplers for household and similar general purposes
IEC 60695-11-10: 1999	Fire hazard testing - Part 11-10: Test flames - 50W horizontal and vertical flame test methods
IEC 60730 (All parts)	Automatic electrical controls for household and similar use
IEC 60825-1 : 1993	Safety of laser products - Part 1: Equipment classification, requirements and user's guide 2
IEC 60885-1 : 1987	Electrical test methods for electric cables - Part 1: Electrical tests for cables, cords and wires for voltages up to and including 450/750 V
IEC 60906 (All parts)	IEC system of plugs and socket-outlets for household and similar purposes
IEC 60990 : 1999	Methods of measurement of touch current and protective conductor current
IEC 60998-2-2: 2002	Connecting devices for low-voltage circuits for household and similar purposes - Part 2-2: Particular requirements for connecting devices as separate entities with screwless-type clamping units

IEC 60999-1 : 1999	Connecting devices - Electrical copper conductors - Safety requirements for screw-type and screwless-type clamping units - Part 1: General requirements and particular requirements for clamping units for conductors from 0.2 mm sq up to 35 mm sq (included)
IEC 61032: 1997	Protection of persons and equipment by enclosures- Probes for verification
IEC 61058-1 : 2000	Switches for appliances - Part 1: General requirements
IEC/TR2 61149: 1995	Guide for safe handling and operation of mobile radio
IEC 61293 : 1994	Marking of electrical equipment with ratings related to electrical supply Safety requirements
IEC 61558-1 : 1997	Safety of power transformers, power supply units and similar - Part 1: General requirements and tests
IEC 61558-2-17: 1997	Safety of power transformers, power supply units and similar- Part 2-17: Particular requirements for transformers for switch mode power supplies
IEC 61965 : 2003	Mechanical safety of cathode ray tubes
IEC 62151 : 2000	Safety of equipment electrically connected to a telecommunication network
IEC Guide 104: 1997	The preparation of safety publications and the use of basic safety publications and group safety publications
ISO 7000 : 1989	Graphical symbols for use on equipment - Index and synopsis
ITU-T Recommendation K17: 1988	Tests on power-fed repeaters using solid-state devices in order to check the arrangements for protection from external interference
ITU-T Recommendation K21 : 1996	Resistibility of telecommunication equipment installed in customers premises to overvoltages and overcurrents

Only the English language text in the International Standard has been retained while adopting it in this Indian Standard, and as such the page numbers are not the same as in the IEC Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Labelling

Language

In which language must the information on the product or on the packaging be indicated? If English is sufficient please indicate NA.

Note: Generally All the particulars and Labelling are made in English Language.

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

9. Manner in which declaration shall be made

4) The particulars of the declarations required to be specified under this rule on a package shall either be in Hindi in Devanagiri script or in English; Provided...

Mandatory labelling on product

What are the mandatory labelling on the product ?

Refer to question of "Mandatory Standards" in the Technical Section.

Mandatory labelling on packaging

What are the mandatory labelling on the packaging ?

Refer to question of "Mandatory Standards" in the Technical Section.

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

4. Regulation for pre-packing and sale etc., of commodities in packaged form.-... no person shall pre-pack or cause or permit to be pre-packed any commodity for sale, distribution or delivery unless the package in which the commodity is pre-packed bears thereon, or on a label is securely affixed thereto, such declarations as are required to be made under these rules.

6. Declarations to be made on every package- All product packages, to which the Packaging Rules apply, are required to bear certain declarations on their principal display panel.

1.
 1. The name and address of the manufacturer, or where the manufacturer is not packer, the name and address of the manufacturer and packer and for any imported package the name and address of the importer shall be mentioned.
 2. The common or generic names of the commodity contained in the package and in case of packages with more than one product, the name and number or quantity of each product shall be mentioned on the package.
 3. The net quantity, in terms of the standard unit of weight or measure, of the commodity contained in the package or where the commodity is packed or sold by number, the number of the commodity contained in the package shall be mentioned.
 4. The month and year in which the commodity is manufactured or pre-packed or imported shall be mentioned in the package...
 5. Retail sale price – The retail sale price to be mentioned on the package must necessarily be the maximum retail price inclusive of all taxes. The price in Indian rupees and paise should be rounded off to the nearest rupee or 50 paise.
 6. Country - It is now mandatory to declare on the product package, the name of the country of origin or manufacture or assembly in case of imported products.

7. No dual MRP - The Amendment prohibits provision of dual maximum retail price (MRP) on the products. Accordingly, no manufacturer, packer or importer is permitted to declare different MRP's on identical pre-packaged commodities unless otherwise permitted.
8. Every package shall bear the name, address, telephone number, E-mail address, if applicable, of the person who can be or the office which can be, contacted, in case of consumer complaints.
9. It shall not be permissible to affix individual stickers on the package for altering or making declaration required under these rules: Provided that for reducing the Maximum Retail Price (MRP), a sticker with the revised lower MRP (inclusive of all taxes) may be affixed and the same shall not cover the MRP declaration made by the manufacturer or the packer, as the case may be, on the label of the package.
10. It shall be permissible to use stickers for making any declaration other than the declaration required to be made under these rules.
11. Where a commodity consists of a number of components and these components are to be packed in two or more units, for sale as a single commodity, the declaration required to be made under sub-rule (1) shall appear on the main package and such package shall also carry information about the other accompanying packages or such declaration may be given on individual packages and intimation to that effect may be given on the main package and if the components are sold as spare parts, all declarations shall be given on each package.

7. Principal display panel its area, size and letter etc :

(2) (i) TABLE-II

Minimum Height of Numeral

Serial Number	Net quantity in weight / volume		Minimum Height in mm
		Normal Case	When blown, formed, molded, embossed or perforated on container
1	Upto 100 cm sq.	1	2
2	Above 100 cm sq. and upto 5002 cm sq.		4
3	Above 500 cm sq. and upto 2500 cm sq.	4	6
4	Above 2500 cm sq.	6	6

(3) The height of letters in the declaration shall not be less than 1 mm height and when blown, formed, molded, embossed or perforated, the height of letters shall not be less than 2 mm.

Provided that the width of the letter or numeral shall not be less than one third of its height, except in the case of numeral '1' and letters (i), (l) and (I);

(4) the sub-rule (1)-(3) shall not apply if the information to be specified on such package under this rule is also required to be given by or under any other law for the time being in force.

8. Declaration Where to appear:- (1) Every declaration required to be made under these rules shall appear on the principal display panel:

Provided that the area surrounding the quantity declaration shall be free from printed information.

1. Above and below by a space equal to at least the height of the numeral in the declaration, and
2. To the left and right by a space at least twice the height of the numeral in the declaration.

9. Manner in which declaration shall be made;- (1) Every declaration which is required to be made on a package under these rules shall be-

(a) Legible and prominent;

(b) numerals of the retail sale price and net quantity declaration shall be printed, printed or inscribed on the package in a colour that contrasts conspicuously with the background of the label....

(3) Where a package is provided with an outside container or wrapper such container or wrapper shall also contain all the declarations which are required to appear on the package except where such container or wrapper itself is transparent and the declarations on the package itself are easily readable through such outside container or wrapper; provided...

Mandatory labelling on the instructions manual

What are the mandatory labelling on the instructions manual (if one is required) ?
In which format, do we have to deliver the instruction manual to our customers ?

Refer to question of "Mandatory Standards" in the Technical Section.

Warnings - General ones

Are general warnings required ? If yes, which ones ?

Refer to question of "Mandatory Standards" in the Technical Section.

Warnings - Specific ones

Are specific warnings required ? If yes which ones ?

Refer to question of "Mandatory Standards" in the Technical Section.

Target Age

Is there any target age applicable? If so, does it have to be indicated and in which form?

Not Applicable

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

20. Action to be taken on completion of inspection of packages at the premises of the manufacturer or the packer:- (1) If it appears from the report referred to in sub-rule (3) of rule 19 that,-

(a) the statistical average of the net quantity contained in the packages drawn as samples under that rule is lesser than the quantity declared on the packages or on the labels affixed thereto, or any such package shows an error in deficiency greater than the maximum permissible error, or

(b) any such package does not bear thereon or on label affixed thereto the declarations to be made under these rules, the Director, Controller or any Legal Metrology Officer shall take action as given below, namely:-

(i) seize the packages drawn by him as samples and shall take adequate steps for the safe custody of seized packages until they are produced in the appropriate court as evidence;

(ii) based on the evidence initiate action for violations of the provisions of the Act and these rules:

Provided that no such action shall be taken if fresh tests are carried out under sub-rule (4) of rule 19, but if after such fresh tests any error or omission as is referred to in this sub-rule is detected, the Director, Controller or any Legal Metrology Officer shall take appropriate action as specified in this sub-rule in accordance with the provisions of the Act against the manufacturer or, as the case may be, the packer.

(3) The disposal of the seized packages shall be made in accordance with the provisions of the Code of Criminal Procedure, 1973 (2 of 1974).

Chapter VII GENERAL

32. Penalty for contravention of Rules.-(1) Whoever contravenes the provisions of rules 27 to 31, he shall be punished with fine of four thousand rupees.

(2) Whoever contravenes any other provision of these rules, for the contravention of which no punishment has been provided either in the Act or in the rules, he shall be punished with fine of two thousand rupees.

Comments

Comments

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

18. Provisions relating to wholesale dealer and retail dealers: (Refer to authority attached)

19. Inspection of quantity and error in packages at the premises of the manufacturer or packer:- (Refer to authority attached)

21. Inspection of quantity and error in packages at the premises of the wholesale dealer or retail dealer :- (Refer to authority attached)

23. Deceptive packages to be repacked or in default to be seized: (Refer to authority attached)

Chapter IV EXEMPTIONS

26. Exemption in respect of certain packages: - Nothing contained in these rules shall apply to any package containing a commodity if-

(a) the net weight or measurement of the commodity is ten gram or ten milliliter or less, if sold by weight or measurement.(**)

Chapter VI REGISTRATION OF MANUFACTURERS, PACKERS AND IMPORTERS

27. Registration of manufacturers, Packers and importers :(Refer to authority attached)

28. Registration of shorter address permissible :.. (Refer to authority attached)

29. Registration of manufacturers and packers, etc.: .. (Refer to authority attached)

Chapter VII GENERAL

31. (1) Any advertisement mentioning the retail ale price of the pre-packaged commodity shall contain a declaration as to the net quantity or number of the commodity contained in the package.

(2) The font size of the net quantity in the advertisement shall be same as that of retail sale price.

30. Compilation of lists of manufacturers or packers and their circulation...

33. Repeal and savings ...

Environment

Recycling - Eco taxes

Which regulations concerning ecodesign and recycling apply to this product? Are there any ecotaxes applicable to this category of product?

Refer to question of “Mandatory Standards” in the Technical Section.

Labelling

What are the mandatory labellings to put on the product or on the packaging regarding the environmental rules ?

Not Applicable

Technical rules / Standards

General safety requirements

What are the general safety requirements ? Does a risk assessment have to be conducted ? If yes, does it concern specific risk ?

Refer to question of “Mandatory Standards” in the Technical Section.

Mandatory or voluntary standards

Are there some mandatory or voluntary standards applicable ? If yes, which ones ?

Please indicate the reference of the applicable standards

IS 616 (2010): Audio, Video and Similar Electronic Apparatus - Safety Requirements [LITD 7: Audio, Video and Multimedia Systems and Equipment]

INTRODUCTION

Principles of safety

General

This introduction is intended to provide an appreciation of the principles on which the requirements of this standard are based. Such an understanding is essential in order that safe apparatus can be designed and manufactured.

The requirements of this standard are intended to provide protection to persons as well as to the surroundings of the apparatus.

Attention is drawn to the principle that the requirements, which are standardized, are the minimum considered necessary to establish a satisfactory level of safety.

Further development in techniques and technologies may entail the need for future modification of this standard.

NOTE : The expression "protection to the surroundings of the apparatus" implies that this protection should also include protection of the natural environment in which the apparatus is intended to be used, taking into account the life cycle of the apparatus, i.e. manufacturing; use, maintenance, disposal and possible end-of-life recycling of parts of the apparatus.

Hazards

The application of this standard is intended to prevent injury or damage due to the following hazards:

- electric shock;
- excessive temperatures;
- radiation;
- implosion;
- mechanical hazards;
- fire.

Electric shock

Electric shock is due to current passing through the human body. Currents of the order of a milliampere can cause a reaction in persons in good health and may cause secondary risks due to involuntary reaction. Higher currents can have more damaging effects. Voltages below certain limits are generally regarded as not dangerous under specified conditions. In order to provide protection against the possibility of higher voltages appearing on parts which may be touched or handled, such parts are either earthed or adequately insulated.

For parts which can be touched, two levels of protection are normally provided to prevent electric shock caused by a fault. Thus a single fault and any consequential faults will not create a hazard. The provision of additional protective measures, such as supplementary insulation or protective earthing, is not considered a substitute for, or a relief from, properly designed basic insulation.

Cause	Prevention
Contacts with parts normally at hazardous voltage.	Prevent access to parts at hazardous voltage by fixed or locked covers, interlocks. etc. Discharge capacitors at hazardous voltages.
Bare breakdown of insulation between parts normally at hazardous voltage and " accessible parts.	Either use double or reinforced insulation between parts normally at hazardous voltages and accessible parts so that breakdown is not likely to occur, or connect accessible conductive parts to protective earth so that the voltage which can develop is limited to a safe value. The insulations shall have adequate mechanical and electrical strength.
Breakdown of insulation between parts normally at hazardous voltage and circuits normally at non-hazardous voltages. there-ther by double or reinforced insulation so that breakdown is not likely to occur, or by a protective earthed screen, or connect the circuit normally at non-hazardous voltage to protective earth, so that the voltage which can develop is limited to a safe value.	Segregate hazardous and non-hazardous voltage circuits ei-
Touch current from parts at hazardous voltage through the human body. (Touch current can include current due to RFI filter components connected between mains supply circuits and accessible parts or terminals.)	Limit touch current to a safe value or provide a protective earthing connection to the accessible parts.

Excessive temperatures

Requirements are included to prevent injury due to excessive temperatures of accessible parts, to prevent damaging of insulation due to excessive internal temperatures, and to prevent mechanical instability due to excessive temperatures developed inside the apparatus.

Radiation

Requirements are included to prevent injury due to excessive energy levels of ionizing and laser radiation, for example by limiting the radiation to non-hazardous values.

Implosion

Requirements are included to prevent injury due to implosion of picture tubes.

Mechanical hazards

Requirements are included to ensure that the apparatus and its parts have adequate mechanical strength and stability, to avoid the presence of sharp edges and to provide guarding or interlocking of dangerous moving parts.

Fire

A fire can result from

- overloads;

- component failure;
- insulation breakdown;
- bad connections;
- arcing.

Requirements are included to prevent any fire which originates within the apparatus from spreading beyond the immediate vicinity of the source of the fire or from causing damage to the surroundings of the apparatus.

The following preventive measures are recommended:

- the use of suitable components and subassemblies;
- the avoidance of excessive temperatures which might cause ignition under normal or fault conditions;
- the use of measures to eliminate potential ignition sources such as inadequate contacts, bad connections, interruptions;
- the limitation of the quantity of combustible material used;
- the control of the position of combustible materials in relation to potential ignition sources;
- the use of materials with high resistance to fire in the vicinity of potential ignition sources;
- the use of encapsulation or barriers to limit the spread of fire within the apparatus;
- the use of suitable fire retardant materials for the enclosure.

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1 General

1.1 Scope

1.1.1 This International Safety Standard applies to electronic apparatus designed to be fed from the MAINS, from a SUPPLY APPARATUS, from batteries or from REMOTE POWER FEEDING and intended for reception, generation, recording or reproduction respectively of audio, video and associated signals. It also applies to apparatus designed to be used exclusively in combination with the above-mentioned apparatus.

This standard primarily concerns apparatus intended for household and similar general use but which may also be used in places of public assembly such as schools, theatres, places of worship and the workplace. PROFESSIONAL APPARATUS intended for use as described above is also covered unless falling specifically within the scope of other standards.

This standard concerns only safety aspects of the above apparatus; it does not concern other matters, such as style or performance.

This standard applies to the above-mentioned apparatus, if designed to be connected to the TELECOMMUNICATION NETWORK or similar network, for example by means of an integrated modem.

Some examples of apparatus within the scope of this standard are:

- receiving apparatus and amplifiers for sound and/or vision;
- independent LOAD TRANSDUCERS and SOURCE TRANSDUCERS;
- SUPPLY APPARATUS intended to supply other apparatus covered by the scope of this standard;
- ELECTRONIC MUSICAL INSTRUMENTS, and electronic accessories such as rhythm generators, tone generators, music tuners and the like for use with electronic or non-electronic musical instruments;
- audio and/or video educational apparatus;
- video projectors;

NOTE 1 Film projectors, slide projectors, overhead projectors are covered by IEC 60335-2-56

- video cameras and video monitors;
- video games and flipper games;

NOTE 2 Video and flipper games for commercial use are covered by IEC 60335-2-82

- juke boxes;
- electronic gaming and scoring machines;

NOTE 3 Electronic gaming and scoring machines for commercial use are covered by IEC 60335-2-82

- teletext equipment;
- record and optical disc players;
- tape and optical disc recorders;
- antenna signal converters and amplifiers;
- antenna positioners;
- Citizen's Band apparatus;
- apparatus for IMAGERY;
- electronic light effect apparatus;
- apparatus for use in alarm systems;
- intercommunication apparatus, using low voltage MAINS as the transmission medium;
- cable head-end receivers;

- multimedia apparatus;

NOTE 4 The requirements of IEC 60950 may also be used to meet the requirements for safety of multi media apparatus (see also IEC Guide 112)

- professional general use amplifiers, record or disc players, tape players, recorders, and public address systems;
- professional sound/video systems;
- electronic flash apparatus for photographic purposes (see Annex L).

1.1.2 This standard applies to apparatus with a RATED SUPPLY VOLTAGE not exceeding

- 250 V a.c. single phase or d.c. supply;
- 433 V a.c. in the case of apparatus for connection to a supply other than single-phase.

1.1.3 This standard applies to apparatus for use at altitudes not exceeding 2 000 m above sea level, primarily in dry locations and in regions with moderate or tropical climates.

For apparatus with protection against splashing water, additional requirements are given in annex A.

For apparatus to be connected to TELECOMMUNICATION NETWORKS, additional requirements are given in annex B.

For apparatus intended to be used in vehicles, ships or aircraft, or at altitudes exceeding 2 000 m above sea level, additional requirements may be necessary.

NOTE See Table A.2 of IEC 60664-1.

Requirements, additional to those specified in this standard, may be necessary for apparatus intended for special conditions of use.

1.1.4 For apparatus designed to be fed from the MAINS, this standard applies to apparatus intended to be connected to a MAINS supply with transient overvoltages not exceeding overvoltage category II according to IEC 60664-1.

For apparatus subject to transient overvoltages exceeding those for overvoltage category II, additional protection may be necessary in the MAINS supply of the apparatus.

1.2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60027 (all parts), Letter symbols to be used in electrical technology

IEC 60038:1983, IEC standard voltages

Amendment 1 (1994)

Amendment 2 (1997)

IEC 60068-2-6:1995, Environmental testing - Part 2: Tests - Test Fc: Vibration (sinusoidal)

IEC 60068-2-32:1975, Environmental testing - Part 2: Tests - Test Ed: Free fall (Procedure 2)

IEC 60068-2-75:1997, Environmental testing - Part 2-75: Tests,- Test Eh: Hammer tests

IEC 60068-2-78:2001, Environmental testing - Part 2: Tests - Test Gab: Damp heat, steady state '.

IEC 60085:2004, Thermal evaluation and classification of electrical insulation

IEC 60086-4:2000, Primary batteries - Part 4: Safety of lithium batteries

IEC 60112:2003, Method for determining the comparative and the proof tracking indices of solid insulating materials under moist conditions

IEC 60127 (all parts), Miniature fuses

IEC 60167:1964, Methods of test for the determination of the insulation resistance of solid insulating materials

IEC 60216 (all parts), Guide for the determination of thermal endurance properties of electrical insulating materials

IEC 60227 (all parts), Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V

IEC 60245 (all parts), Rubber insulated cables - Rated voltages up to and including 450/750 V

IEC 60249-2 (all specifications), Base materials for printed circuits - Part 2: Specifications

IEC 60268-1:1985, Sound system equipment - Part 1: General

IEC 60317 (all parts), Specifications for particular types of winding wires

IEC 60320 (all parts), Appliance couplers for household and similar general purposes

IEC 60335-1:2001, Household and similar electrical appliances - Safety - Part 1: General requirements Amendment 1 (2004)

IEC 60384-1:1999, Fixed capacitors for use in electronic equipment - Part 1: Generic specification

IEC 60384-14:1993, Fixed capacitors for use in electronic equipment - Part 14: Sectional specification: Fixed capacitors for electromagnetic interference suppression and connection to the supply mains

Amendment 1 (1995)

IEC 60417 (all parts), Graphical symbols for use on equipment

IEC 60454 (all parts), Specifications for pressure-sensitive adhesive tapes for electrical purposes

IEC 60529:1989, Degrees of protection provided by enclosures (IP Code)

Amendment 1 (1999)

IEC 60664-1:1992, Insulation coordination for equipment within low-voltage systems - Part 1: Principles, requirements and tests

Amendment 1 (2000)

Amendment 2 (2002)

IEC 60664-3:2003, Insulation coordination for equipment within low-voltage systems - Part 3: Use of coatings to achieve insulation coordination of printed board assemblies

IEC 60691 :2002, Thermal links - Requirements and application guide

IEC 60695-2-2:1991, Fire hazard testing - Part 2: Test methods - Section 2: Needle-flame test

IEC 60695-11-10:1999, Fire hazard testing - Part 11-10: Test flames - 50 W horizontal and vertical flame test methods
Amendment 1 (2003)

IEC 60707:1999, Flammability of solid non-metallic materials when exposed to flame sources List of test methods

IEC 60730 (all parts), Automatic electrical controls for household and similar use

IEC 60825-1 :1993, Safety of laser products - Part 1: Equipment classification, requirements and user's guide 2

Amendment 1 (1997)

Amendment 2 (2001)

IEG 60851-3:1996, Methods of test for winding wires - Part 3: Mechanical properties

Amendment 1 (1997)

IEC 60851-5:1996, Methods of test for winding wires - Part 5: Electrical properties

Amendment 1 (1997)

Amendment 2 (2004)

IEC 60851-6:1996, Methods of test for winding wires - Part 6: Thermal properties

IEC 60884 (all parts), Plugs and socket-outlets for household and similar purposes

IEC 60885-1:1987, Electrical test methods for electric cables - Part 1: Electrical tests for cables, cords and wires for voltages up to and including 450/750 V

IEC 60906 (all parts), IEC system of plugs and socket-outlets for household and similar purposes

IEC 60950:1999, Safety of information technology equipment

IEC 60990:1999, Methods of measurement of touch current and protective conductor current

IEC 60998-2-2:2002, Connecting devices for low-voltage circuits for household and similar purposes - Part 2-2: Particular requirements for connecting devices as separate entities with screwless-type clamping units

IEC 60999-1:1999, Connecting devices - Electrical copper conductors - Safety requirements for screw-type and screwless-type clamping units - Part 1: General requirements and particular requirements for clamping units for conductors from 0,2 mm sq up to 35 mm sq (included)

IEC 61032:1997, Protection of persons and equipment by enclosures - Probes for verification

IEC 61051-2:1991, Varistors for use in electronic equipment - Part 2: Sectional specification for surge suppression varistors

IEC 61058-1 :2000, Switches for appliances - Part 1: General requirements

IEC/TR2 61149:1995, Guide for safe handling and operation of mobile radio equipment

IEC 61260:1995, Electroacoustics - Octave-band and fractional-octave-band filters

IEC 61293:1994, Marking of electrical equipment with ratings related to electrical supply Safety requirements

IEC 61558-1:1997, Safety of power transformers, power supply units and similar - Part 1: General requirements and tests *

Amendment 1 (1998)

IEC 61558-2-17:1997, Safety of power transformers, power supply units and similar - Part 2-17: Particular requirements for transformers for switch mode power supplies

IEC 61965:2003, Mechanical safety of cathode ray tubes

IEC 62151 :2000, Safety of equipment electrically connected to a telecommunication network

IEC Guide 104:1997, The preparation of safety publications and the use of basic safety publications and group safety publications

ISO 261 :1973, ISO general purpose metric screw threads - General plan

ISO 262:1973, ISO general-purpose metric screw threads - Selected sizes for screws, bolts and nuts

ISO 306:1994, Plastics - Thermoplastic materials - Determination of Vicat softening temperature (VST) .

ISO 7000:1989, Graphical symbols for use on equipment - Index and synopsis

ITU-T Recommendation K17:1988, Tests on power-fed repeaters using solid-state devices in order to check the arrangements for protection from external interference

ITU-T Recommendation K21:1996, Resistibility of telecommunication equipment installed in customer's premises to overvoltages and overcurrents

2 Definitions

For the purpose of this International Standard, the following definitions apply.

(Please refer to the PDF File attached)

3 General requirements (Please refer to the PDF File attached)

4 General test conditions (Please refer to the PDF File attached)

5 Marking and instructions (Please refer to the PDF File attached)

6 Hazardous radiations (Please refer to the PDF File attached)

7 Heating under normal operating conditions (Please refer to the PDF File attached)

8 Constructional requirements with regard to the protection against electric shock (Please refer to the PDF File attached)

9 Electric shock hazard under normal operating conditions (Please refer to the PDF File attached)

10 Insulation requirements (Please refer to the PDF File attached)

11 Fault conditions (Please refer to the PDF File attached)

12 Mechanical strength (Please refer to the PDF File attached)

13 CLEARANCES and CREEPAGE DISTANCES (Please refer to the PDF File attached)

14 Components (Please refer to the PDF File attached)

15 TERMINALS (Please refer to the PDF File attached)

16 External flexible cords (Please refer to the PDF File attached)

17 Electrical connections and mechanical fixings (Please refer to the PDF File attached)

18 Mechanical strength of picture tubes and protection against the effects of implosion (Please refer to the PDF File attached)

19 Stability and mechanical hazards (Please refer to the PDF File attached)

20 Resistance to fire (Please refer to the PDF File attached)

ANNEX A to ANNEX N (Please refer to the PDF File attached)

Physical and mechanicals properties

What are the physical and mechanicals properties required ?

Refer to question of “Mandatory Standards” in the Technical Section.

Electrical aspects

What are requirements regarding electrical aspects ?

Refer to question of “Mandatory Standards” in the Technical Section.

Inflammability

Are there any specific requirements to ensure inflammability?

Refer to question of “Mandatory Standards” in the Technical Section.

Hygiene

What are the rules regarding hygiene ?

Not Applicable

Radioactivity

What are the rules regarding radioactivity ?

Refer to question of “Mandatory Standards” in the Technical Section.

Radiofrequency (2G,3G,4G,5G, Bluetooth, Ant+, GPS, Wifi...)

What are the rules regarding the radiofrequency?

Products making noise

What are the requirements regarding products making noise ?

Not Applicable

Chemicals substances

What are the requirements regarding chemicals substances ? Which ones are forbidden ? Limited ? Do we have to clearly indicate all substances that are in the product, or is the indication of major substances enough?”

Not Applicable

Packaging requirements

Can you please indicate if there are some requirements regarding the packaging of our products (mainly they are in paper, carton, plastic, textile) ? If yes, please explain them.

Not Applicable

Laboratory

In which laboratory can we perform the testing of the product according to the standard?

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Group 2 Lab of National Repute and Eminence, Facilities

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Not Applicable

Comments

Comments

Certifications / Registrations

Certifications or registrations

Are there some certifications or registrations applicable to the product ?

Voluntary ISI Certificate may be procured.

Laboratory

In case of mandatory certification/ registration that we must follow please precise the name and address of any laboratories where we can conduct them

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Group 2 Lab of National Repute and Eminence, Facilities

Comments

Comments

The steps of getting ISI Certificate are as below -:

Step 1 : Confirmation of Product Quality as per concerned IS Standard

- Identification of ISI Standard Code
- Collection of the Testing Sample from applicant
- Testing of sample at our lab
- Issuing the guideline about Non-Conformity, if any
- Issue of Test Report for your reference

Step 2 : Filing of the Application Form

- Arrange the Internal Test Report of product as per test report
- Compiling of all required documents
- Remit the application fees direct to BIS in USD*
- Filling-up the Form-V
- Submission of Application Form with all enclosures
- Get the confirmation about scrutiny of documents by BIS Office

Step 3 : Remittance of Audit Fees & Travelling Expenses for Inspection

- Pay the audit fees for inspection
- Get the confirmation about BIS Inspection Team
- Organize the VISA, Ticket, Lodging & Boarding for visiting team
- Inspection team will visit to Factory for Physical Verification
- Team will verify the Factory Set-up & Quality Control Process
- Team will pick the random sample to be tested in BIS Approved Lab

Step 4 : Organize the Lab Test Report (LTR)

- Select the BIS recognized lab as per guideline of BIS Office
- Provide the sample (picked-up by team) of product to get tested
- Get the test result

Step 5 : Closure of Non-Conformity (if any) and Remit the Official Fees

- Rectify the process of Factory Unit, if inspection team points out
- Submit the proof of rectification done by unit to close non-conformity
- Pay the annual marking fees (annual license fees) to BIS
- Submit the Bank Guarantee to BIS

Step 6 : Get the ISI Marking /Registration Certificate

Conformity documents

Documents

What are the conformity documents ? Can you please list them ?

Not Applicable

For ISI Certificate, refer to comments sections of “Certifications/Registrations”.

Format

Does it exist a specific format for the conformity documents ? If yes, can you please provide us an example in english ?

No Particular Format

Comments

Comments

Importation rules

General rules

What are the general rules in order to import the product ?

Generally, Importers are required to register with the DGFT to obtain an Importer Exporter Code Number (IEC) issued against their Permanent Account Number (PAN), before engaging in EXIM activities. After an IEC has been obtained, the source of items for import must be identified and declared.

IMPORT PROCEDURE

Step1 : File Bill of Entry with Business Identification Number (BIN)

Step 2: Determine rate of duty for clearance from warehouse

Step 3: File requisite documents with customs department

Step 4: Submit import report/ manifest

Step 5: Receive permission to import goods

Documents

What are the required documents ?

Mandatory Documents:

1. Importer Exporter Code issued from the DGFT
2. Bill of Entry
3. Examination Order generated by the EDI system of Customs

Optional Documents depending upon the type of Product being Imported:

1. Certificate of Origin issued by Authorized Person/ Agency at the place of manufacturing/ processing etc. of the food consignment. Certificate of Origin shall contain information on Country of Origin etc. Hence if the consignor is from a different country;
2. Stuffing list, Packing List;
3. Commercial invoice as mentioned in the Bill of Entry (BoE);
4. Bill of Lading as mentioned in the Bill of Entry (BoE) for sea consignment;
5. Air Way Bill as mentioned in the Bill of Entry (BoE) for air consignment;

Time period

In which time period may we obtain the right to import ?
It generally takes 45 to 60 days for completing the overall procedure from import clearance to Product reaching store shelf.

Administration

What is the administration in charge of importations ? Name and address ?

- Directorate General of Foreign Trade
- Central Board of Excise and Customs

Comments

Comments

Inspection of the product

Inspection method

Does it exist an inspection procedure applicable to the product ? If yes, which administration is in charge of ? Can you please describe the procedure ?

Not Applicable

Commercialization rules

Notification / Declaration

What are the general rules in order to comercialize a product ?
Is there a notification / declaration to do ?

Trade license will have to be acquired which is governed by Municipality Act- which will location specific.
Notification is location specific.

Documents

What are the required documents ?

General documents will be required-in relation to establishment of business.

Delay

What is the delay to obtain the right to commercialize ?

Not Applicable

Display in store

Is there some specific rules on the way to display the product in store ?

Not Applicable

Display on internet

Is there some specific mentions to display the product on internet ?

Information to consumers

Do some specific information to have be given to consumers when the product is placed on the market ?

Not Applicable

Comments

Comments

Administration

What is the administration in charge of commercialization ? Name and address ?

Department of Legal Metrology which comes under the Department of Consumer Affairs.

Documents